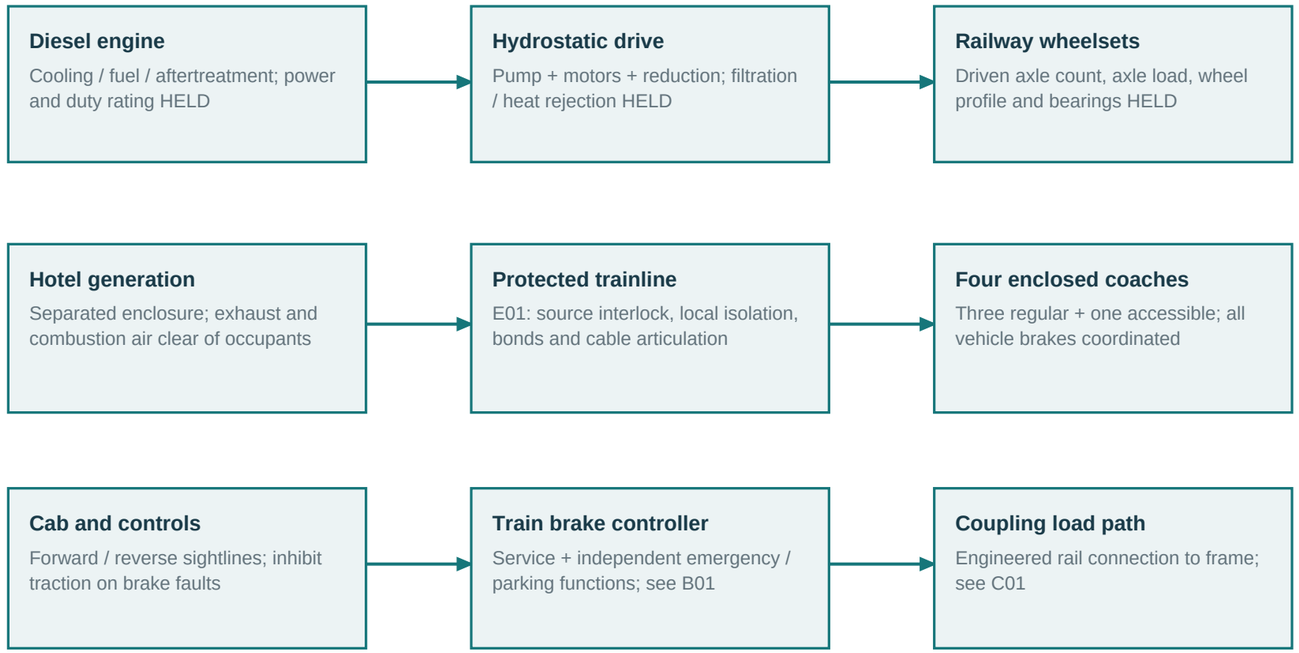
Locomotive and train interface architecture

Selection basis: integrator-supplied diesel-hydrostatic power chassis; steam-era shell is cosmetic. Arrangement is functional, not a measured locomotive.

L1 / PROPULSION, HOTEL POWER AND SAFETY INTERFACES



L2 / DIMENSIONS AND RATINGS TO OBTAIN

HELD: engine model and rated continuous net power; hydrostatic duty / reduction; adhesion by driven wheel load; chassis mass and axle distribution; coupler height; wheel profile; curve and grade limits.

HELD: complete cooling balance at Southern Nevada ambient, service clearance, fuel capacity, exhaust backpressure and hottest-surface segregation.

Obtain the integrator GA before drawing cosmetic boiler, smokebox, cab or tender around working machinery. Do not scale a Jupiter photograph.

L3 / VERIFIED INTERFACES BEFORE PURCHASE

Rated drawbar pull across speed, starting resistance and rail conditions; braking demonstration at gross train mass; continuous thermal capacity at slow speed; towing and rescue method; documented emergency stop and interlocks.

Engine power alone does not prove usable rail tractive effort. The 14,500 kg locomotive + hotel allocation is not all assumed driven mass.

MASS BASIS / STUDY

Nominal 54,755 kg; stress 59,840 kg; upper screen 65,000 kg. The 65 t performance case can exceed current coach frame load cases; if adopted, revise every vehicle / bogie case. These are allocations, not supplier ratings.

OPERATING INPUTS / STUDY

Owner basis: 0% longitudinal track grade. Engineer sets survey acceptance and allowable deviation. 5 / 7.5 mph and 30 / 50 / 75 m radii remain study inputs; these are not operating approvals.

RELEASE GATE / HELD

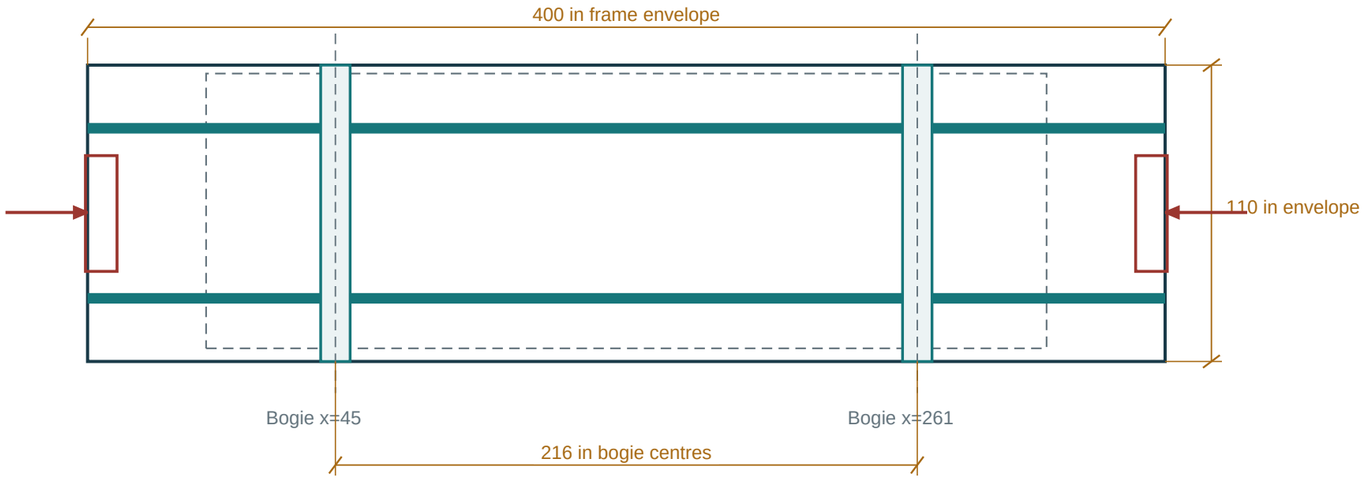
Railway integrator confirms drive, brake, chassis and shell interfaces; system engineer closes mass / power / thermal budget. Refer to PU source register and interface schedule.

VERIFIED = cited source fact | CALCULATED = stated-input arithmetic | STUDY = layout / design basis | HELD = evidence or engineering required

Coach frame load paths and support interfaces

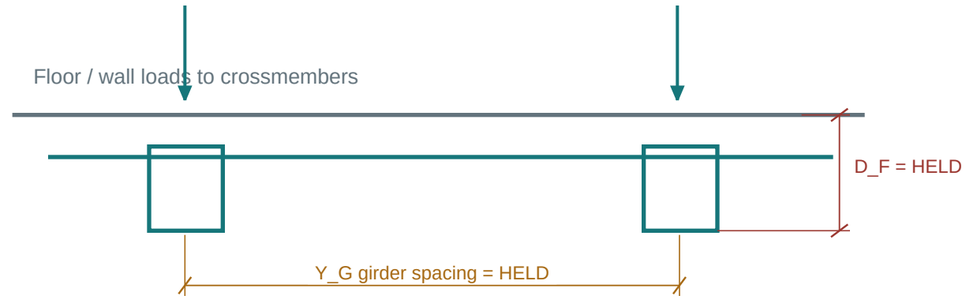
All shown members are symbols. Member sections, depth, spacing, grades, welds, bolts and connection stresses remain HELD.

F1 / FRAME PLAN - DIMENSIONED STUDY ENVELOPE



End load boxes tie couplings into both girders; attachment geometry HELD.

F2 / TRANSVERSE SECTION - SYMBOLIC



Section shapes are not actual profiles.

No structural credit assigned to cabin walls or roof. Outboard floor, wall, stair and lift loads require designed transfer into both girders.

F3 / LONGITUDINAL SUPPORT AND CANTILEVER INTERFACES

Body edge to support: 48 in; platform extension: 44 in each end.

LOAD CASES TO RESOLVE / FR REGISTER

Static balanced reactions are not axle ratings. Regular coach stress total 11,825 kg => 5,912.5 kg per bogie only when symmetric. Front-half crowd screening gives about 7,607 / 4,218 kg before dynamic factors.

A 1 kN tip load produces +1.426 / -0.426 kN support reactions and 2.337 kN-m cantilever moment. Include coupling tension / compression, brake torque, jacking, derailment recovery, lift and stair point loads.

FABRICATION RELEASE HOLD

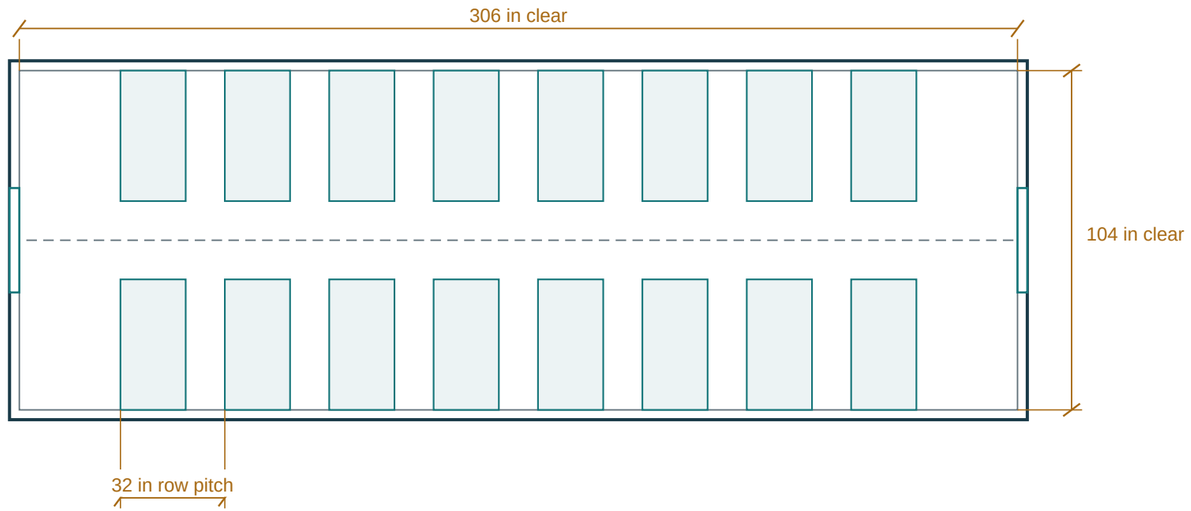
Complete frame torsion, fatigue, buckling and attachment analysis; define weld procedure, qualifications and inspection. Rendered 3 in depth is not a structural selection; 28 in floor height remains provisional. A 65 t train performance case can require a 13.171 t regular-coach allocation, exceeding the current 11.825 t frame case; revise before adoption.

VERIFIED = cited source fact | CALCULATED = stated-input arithmetic | STUDY = layout / design basis | HELD = evidence or engineering required

Regular coach: body interfaces and assembly sequence

Retained envelope: 306 x 104 in clear cabin; 312 x 110 in exterior body; eight rows of four adults. Circulation and emergency egress remain under review.

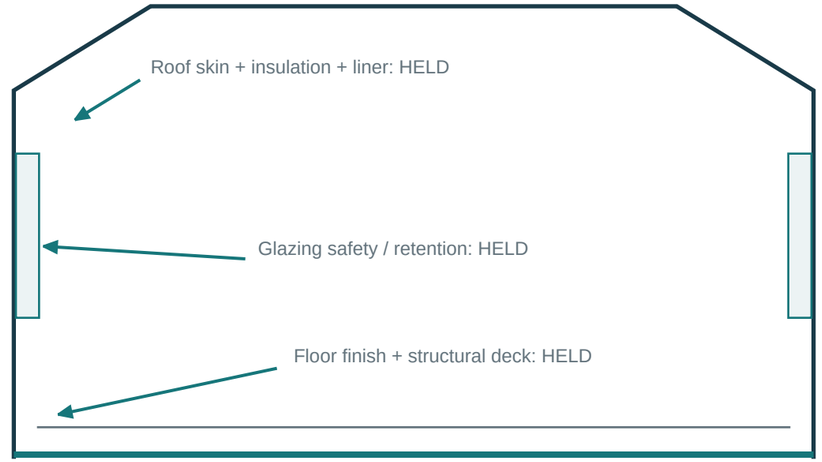
R1 / FLOOR PLAN - SEATING AND BODY INTERFACES



20 in seat depth envelope; 40 in two-adult bench; 24 in aisle.

32 in clear end doors are study values, not an egress approval.

R2 / WALL - FLOOR - ROOF INTERFACE SECTION



Walls attach to frame perimeter; openings need local reinforcement.

No thickness depicted here is a build dimension.

R3 / JOINT DEVELOPMENT

Floor: drained outer edge, sealed penetrations, removable service access away from walking paths.

Wall base: mechanical attachment + corrosion isolation; water path outside floor assembly.

Roof: drained seams, HVAC curb loads to reinforced structure, condensate routed below occupied area.

R4 / BODY ASSEMBLY ORDER

1. Frame acceptance and dimensional survey.
2. Dry-fit deck, walls, end doors and roof.
3. Install protected wiring / HVAC pathways.
4. Water test before insulation and liner.
5. Secure seats to engineered load path.
6. Finish, inspect and weigh each completed car.

R5 / MOCKUP AND RELEASE HOLDS

Full-width mockup with six adults: seat pitch, aisle passing, door operation, window reach and emergency exit route.

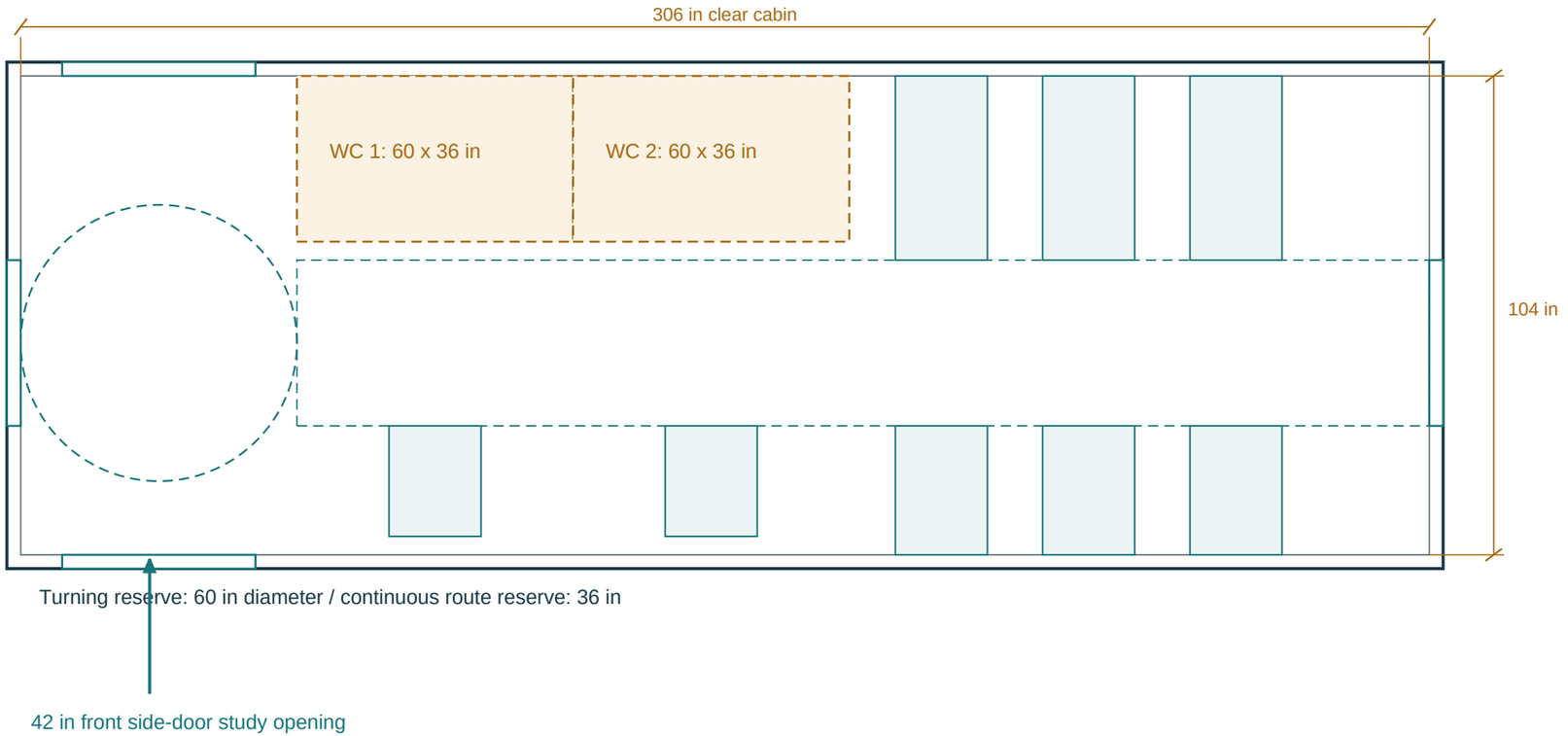
Engineer defines seat anchor loads, floor panels, glazing, fire properties, insulation and door hardware. No ordinary building detail automatically governs the vehicle.

VERIFIED = cited source fact | CALCULATED = stated-input arithmetic | STUDY = layout / design basis | HELD = evidence or engineering required

Accessible coach: circulation, boarding and rescue

Layout reserves are study dimensions. Applicable accessibility basis, securement, lift rating, maneuvering and evacuation require specialist confirmation.

A1 / CLEAR FLOOR PLAN - 11 FIXED SEATS + 2 WHEELCHAIR PLACES



A2 / NORMAL BOARDING

Provide accessible boarding at BOTH stations; North Pole coordinates may follow later. Prefer matched platform + engineered bridge plate at the retained 42 in side door.

Lift alternative is UNINTEGRATED: researched Braun 1,000 lb unit needs 45.85 in minimum clear width / 50.96 in overall width; existing 42 in door does not fit. A 48 in opening study still needs full packaging.

HELD: height, gap / step, device rating, gates, approach and rescue method.

A3 / DISABLED-TRAIN RESCUE

Normal lift failure cannot be the sole evacuation strategy. Establish rescue location, all-weather access, independent lowering / bridging means and staff-assist procedures.

No routine carry-down of occupied wheelchairs. Confirm how each expected mobility device transfers safely at a stalled vehicle.

ANCHORAGE / HELD

Wheelchair location, orientation, securement and occupant restraint depend on the verified regulatory basis and chosen system. Anchor loads transfer into engineered frame nodes, not floor skin alone.

DOOR AND INTERIOR REVIEW

36 in end-door reserve; front side openings 42 in. Verify clear width with hardware, thresholds, trim and handles installed. Protect circulation from seat backs, bags, hinges and equipment.

FULL-SCALE ACCEPTANCE

Mockup with real occupied mobility devices + companion seats; demonstrate entry, turn, park, exit and emergency retrieval. Resolve BB register before frame / body release.

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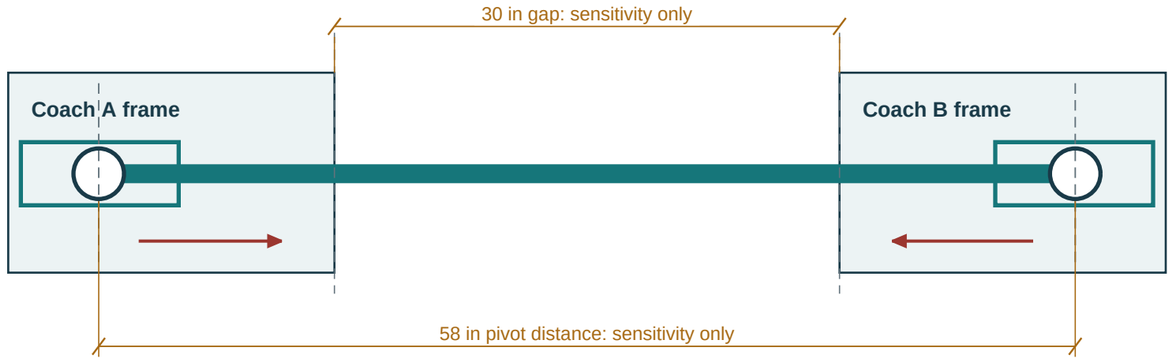
Rail connection: articulated drawbar concept and holds

FUNCTIONAL CONCEPT ONLY - not a copy of any supplier assembly. Choose a documented railway coupling / drawbar system before detailing pins, lugs or welds.

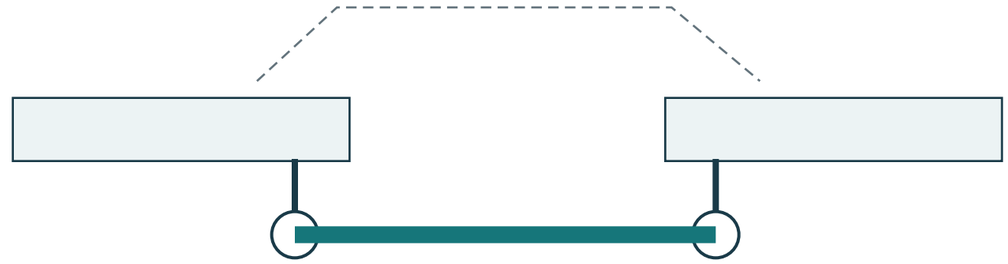
C1 / PLAN - TWO ARTICULATING JOINTS



14 in pin setback each end is retained CAD sensitivity, not hardware selection.

Real rail references: Dellner People Mover semi-permanent / Voith 430-530; fit unproven.

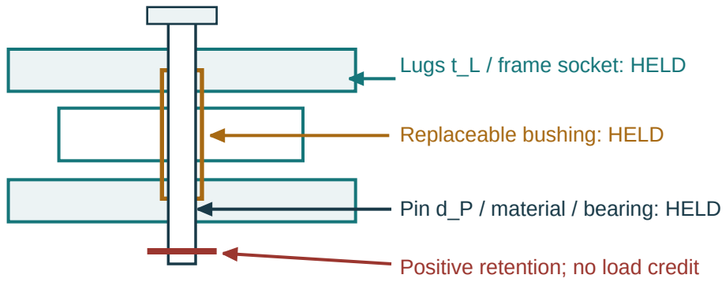
C2 / SIDE - VERTICAL ARTICULATION AND HOSE ROUTE



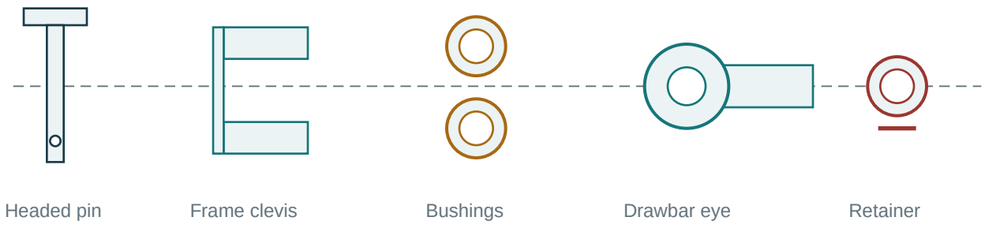
Flexible services routed above / beside protected connection.

Pitch / roll joint arrangement, Z_C height and travel = HELD.

C3 / PIN JOINT SECTION - LOAD-BEARING AND RETENTION FUNCTIONS



C4 / EXPLODED FUNCTIONAL STACK



Functional exploded parts; actual articulation and bearing type HELD.

CALCULATION / RATING HOLD

Resolve tension, braking compression, slack impact, reverse operation, combined yaw / pitch / roll, wear and fatigue. Validate full train reverse-curve kinematics; compare matched railway coupler or engineered captive drawbar. No trailer tongue or generic hitch substitution.

FRAME AND SERVICE INTERFACE HOLD

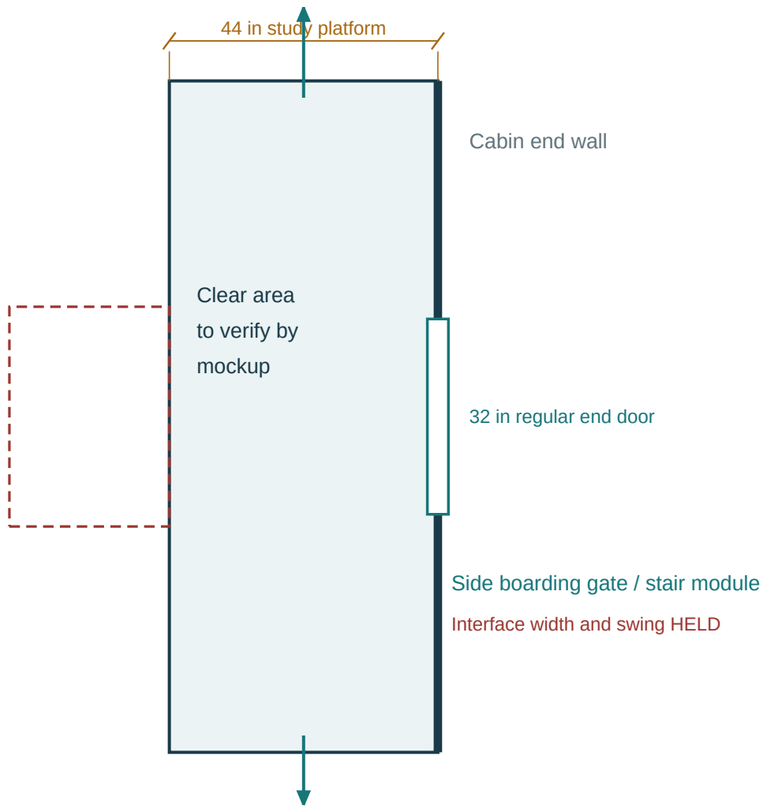
Socket ties into both longitudinal girders through designed diaphragms. Define inspection access, fail-safe retention, wear limits and replaceable parts. Any secondary retention, inter-car guarding, hose loops and disconnect behavior must be coordinated. Refer BC register.

VERIFIED = cited source fact | CALCULATED = stated-input arithmetic | STUDY = layout / design basis | HELD = evidence or engineering required

End platforms, stairs and vehicle-to-station interfaces

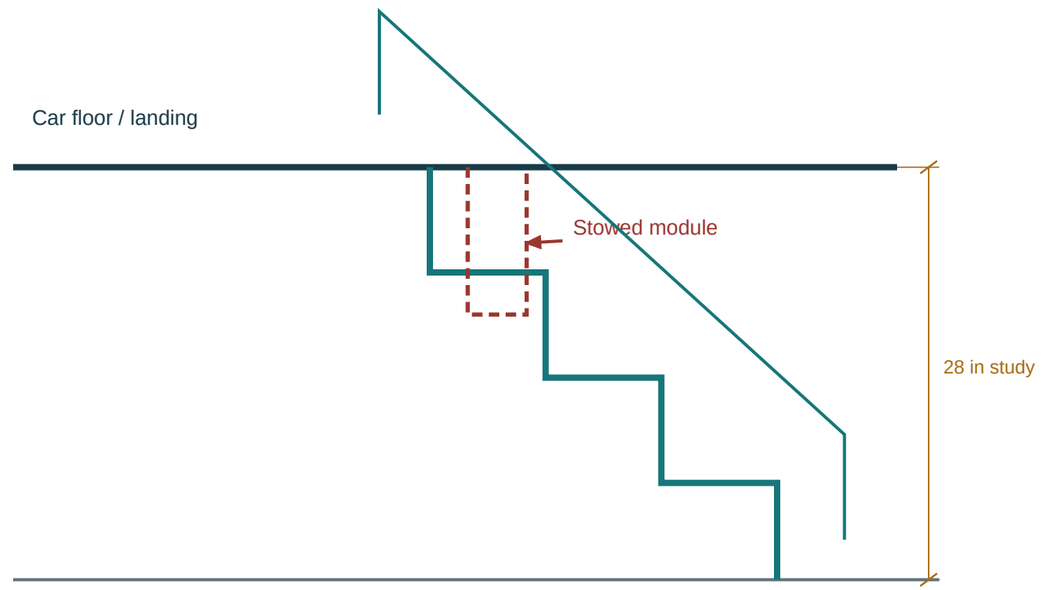
Develop passenger protection around rail connections. The 44 in platform and 28 in floor height are retained packaging studies, not approved boarding geometry.

S1 / END PLATFORM PLAN - MOVING CONNECTION GUARDED



Coupling / pinch envelope below platform: keep enclosed and inaccessible.

S2 / STAIR DEPLOYED AND STOWED - SYMBOLIC SIDE VIEW



Top of rail reference; ground / station surface not fixed

Step count / rise / going shown symbolically; all final values HELD.

Guard and handrail heights, grip, gate latch and interlocks HELD.

S3 / PROTECTED BOARDING

Provide slip resistance, contrast, drainage, lighting and graspable handholds. No access to drawbars, wheel paths or energized services. Gates must stay secure under operation and allow the approved emergency egress method.

S4 / CLEARANCE VERIFICATION

3D swept envelope includes full load / rebound, curve cant, body roll, articulation, wheel wear, track tolerance and stair stowage. Interlock or positively secure stowage before movement; establish inspection method.

S5 / FULL-SCALE TRIAL

Build a non-passenger mockup first. Confirm adult foot placement, reach, door swing, staff position and boarding flow. Accessible boarding uses A01; stair module is not the wheelchair evacuation solution.

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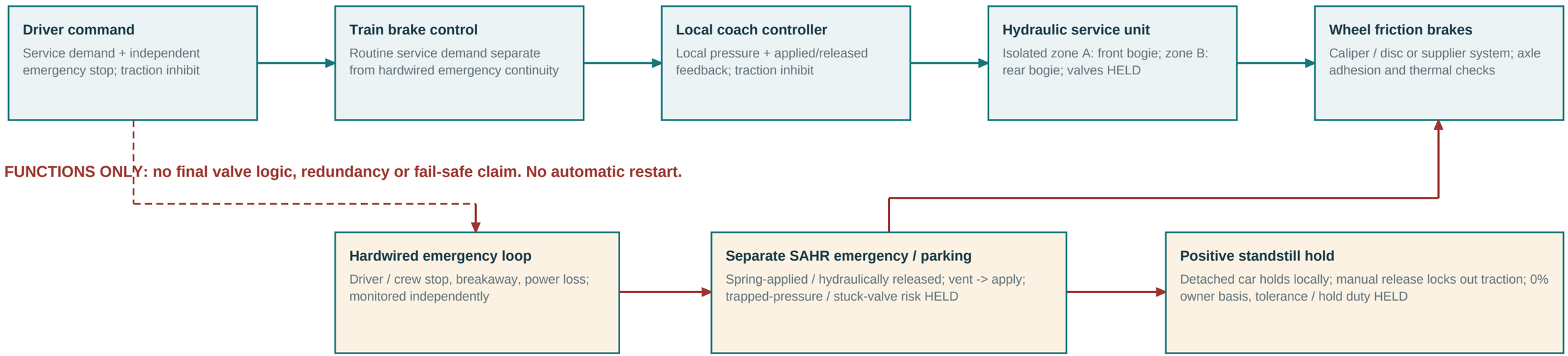
PULSIPHER FARMS / CHRISTMAS RAILWAY

Train brake functional architecture

Owner preference retained: hydraulic service actuation under evaluation. All devices and line styles are functional placeholders; circuit and ratings require brake-system engineering.

B1 / COMMAND, ENERGY AND FAIL-SAFE FUNCTIONS



SERVICE / BALANCE

Design blended response across locomotive and every coach; verify empty / loaded brake balance, wheel slide and low-adhesion effects. Pressure ratings, bore sizes, disc sizes, fluid, hoses and control protocol remain HELD.

FAILURE / SAFETY

Analyze ruptured hose, leaking actuator, loss of electrical power, lost command, separation and common-cause faults. Prove remaining braking / stopping capability under the required fault scenarios. Do not equate release pressure loss with a proven complete emergency function.

TEST AND RELEASE

Dynamometer / supplier thermal proof + loaded stopping tests on surveyed track. Owner basis: 0% longitudinal grade; hold duty includes accepted deviation and fault cases. Test parking, single faults, separation simulation and recovery. Specialist approves the circuit.

ALTERNATIVE TO COMPARE BEFORE SELECTION

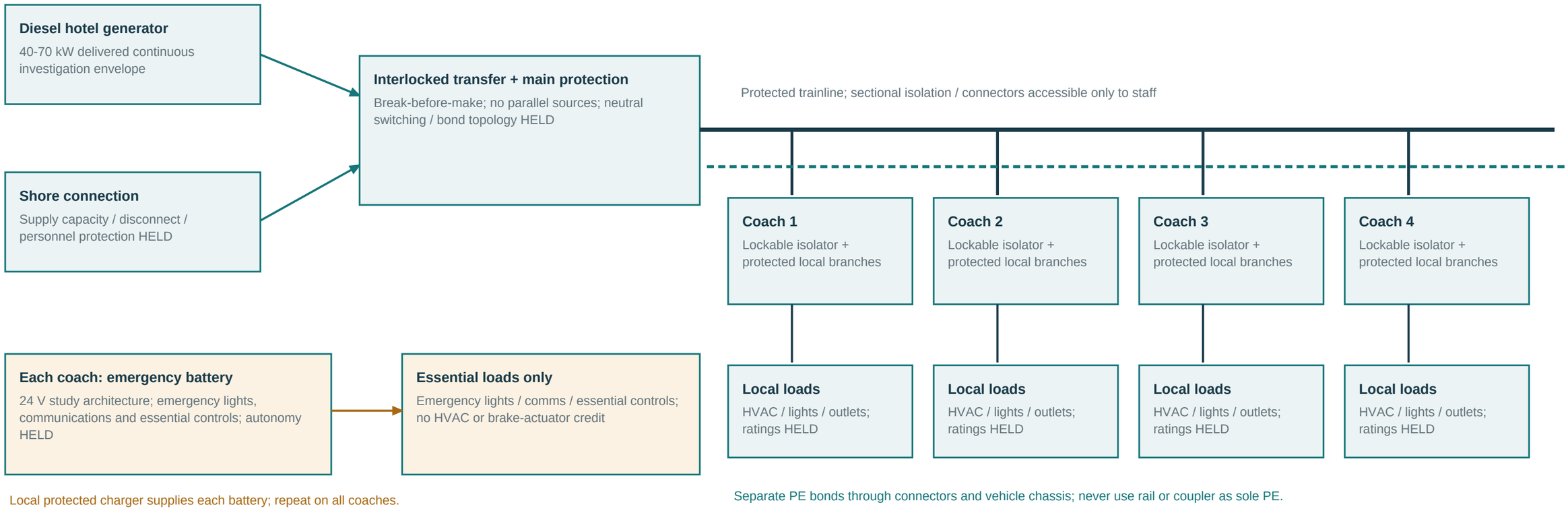
Compare automatic pneumatic train brakes with H1 hydraulic service + separate SAHR emergency / parking. Keep hydraulic preference pending specialist selection. Knorr HydroAct provides a real rail-hydraulic reference, not a selected installation. Required valve redundancy, common-cause separation and zero-energy hold remain HELD. Refer BC register.

VERIFIED = cited source fact | CALCULATED = stated-input arithmetic | STUDY = layout / design basis | HELD = evidence or engineering required

Hotel power one-line and protected distribution

Power topology only: cable, breaker, connector, voltage, frequency, short-circuit capacity, neutral and grounding arrangement are HELD pending equipment selection.

E1 / SOURCE TRANSFER AND TRAINLINE - SINGLE-LINE SCHEMATIC



LOAD BASIS / STUDY

Dry-weather hotel running screen about 27.54 kW; humid / ventilation sensitivity about 35.56 kW at assumed cooling COP 2.5 + 3 kW other loads. These are calculations, not equipment ratings. Heating and starting duty still require selection.

HVAC / SAFETY INTERFACES

Include ambient derating, startup/inrush, heater coincidence, condensate drainage, vibration, intake / exhaust separation and maintenance access. Brake control power is separately coordinated; lighting batteries are not presumed to operate train brakes.

RELEASE HOLD / PU REGISTER

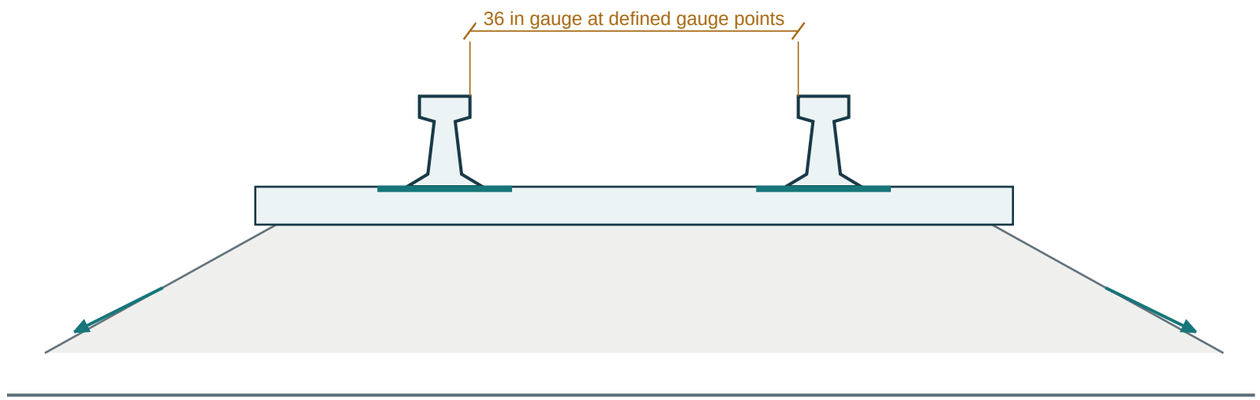
480 V three-phase, 60 Hz is only an option. Licensed electrical design / integrator must establish topology, protective coordination, fault current, cable flex life, bonding and test regime. No conductor size or breaker rating is selected here.

VERIFIED = cited source fact | CALCULATED = stated-input arithmetic | STUDY = layout / design basis | HELD = evidence or engineering required

Track section, survey inputs and station berth interfaces

Owner basis: 0% longitudinal grade. No site alignment is issued; North Pole coordinates remain deferred. Both stations require boarding and evacuation interfaces.

T1 / TRACK SECTION - 36 in GAUGE STUDY BASIS



Prepared formation / subgrade; drainage and geotechnical criteria HELD

Section diagram symbolic: rail profile, tie and ballast dimensions are not selected.

T2 / TRACK DESIGN INPUTS

OWNER BASIS: 0% longitudinal grade. Track engineer defines vertical alignment, survey precision, acceptance tolerance and allowable deviation. Provide drainage independently; level rail does not mean level site grading.

Survey centerline, stations, soils / settlement, crossings, irrigation / flood paths, storage and access. North Pole coordinates remain deferred.

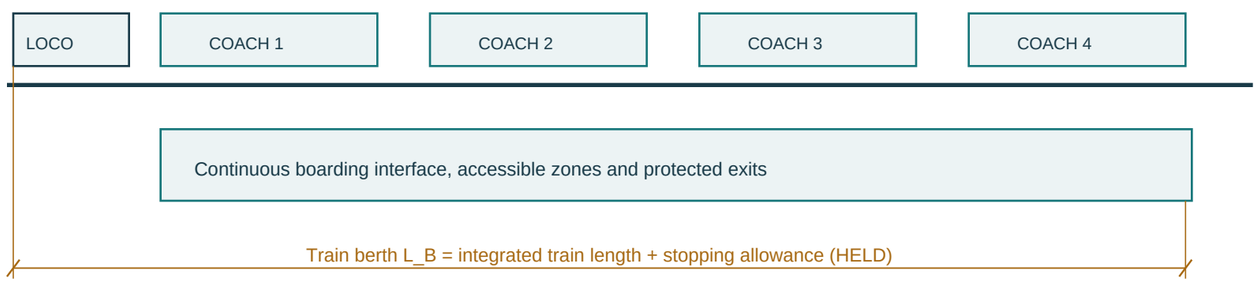
HELD: rail / joints, ties, fastening, spacing, ballast / subgrade, cant, gauge tolerance and wear. Coordinate wheel back-to-back / flange, rail head, check rails and turnouts. Gauge match alone is insufficient.

T3 / FARM CREW / SPECIALIST BOUNDARY

Crew can support survey staking, approved grading / drainage, material handling and supervised tangent assembly after instruction. Specialist establishes first article, curves, transitions, turnouts and inspection standards.

Do not field-bend rail to a guessed radius or use consumer laser level accuracy as track acceptance. Final line / level / gauge acceptance needs the specified instruments and tolerances.

T4 / STATION BERTH ENVELOPE - SITE-NEUTRAL INTERFACE



T5 / VEHICLE CLEARANCE AND DISPATCH

Set platform edge only after dynamic swept-envelope analysis; coordinate provisional 28 in floor with real bogies / suspension. Stations need safe local isolation, clear dispatch sightlines, lighting and rescue access. Defer coordinates, not vehicle-to-station interface decisions. Refer TD register.

VERIFIED = cited source fact | CALCULATED = stated-input arithmetic | STUDY = layout / design basis | HELD = evidence or engineering required